Digital Vector · VCE Applied Computing | Unit 2 · Area of Study 1 · Innovative Solutions · KK02

Functions & capabilities of digital systems

Before you can design something innovative, you need a clear-eyed picture of what today's digital systems already do — and how much they are already able to do. This dot point is the map of the territory you're about to push past.

What this key knowledge covers
This summary distils everything you need for **KK02 — Functions & capabilities of digital systems** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

What counts as a digital system ?

A digital system is never just "the app" or "the gadget". It's a **combination of parts working together** to carry out an information-processing job.

Tarni's contactless payment isn't "the card reader". It's the reader (hardware) **plus** the bank's software, **plus** your account data, **plus** the payment network linking the shop to your bank, **plus** the people and rules that set it up. Pull any one out and the "system" stops being a system.

The physical parts — sensors, chips, readers, satellites, robot arms.

The instructions that tell the hardware what to do and when.

The raw material it acts on — locations, balances, traffic counts.

The links that connect parts, and the humans/processes giving it purpose.

Function vs capability

This is the one idea the whole dot point turns on — and the one students most often blur. Get it crisp and the rest is straightforward.

The task or role the system is designed to perform. Answers **"what's its job?"** — usually one clear verb: locate, pay, listen, lift, control.

The **extent** of what it can achieve — its scale, speed, accuracy, reach or autonomy. Answers **"how much / how well / how far?"** — usually a measure.

Function = your job description: "make coffees, serve customers." **Capability** = what you can actually deliver: "180 cups an hour at peak, latte art, remembers thirty regulars' orders." The job is the same; the capability is what makes you good at it.

Both share one **function** — give off light. Their **capability** is worlds apart: a torch lights a footpath; a lighthouse throws a beam 40 km out to sea, all night, in fog. Same job, different capability.

Flip these to test yourself — the front is a system, the back splits its function from a capability.

Function: locate the device and give directions. **Capability:** pinpoints to within ~3 m, anywhere on Earth, in real time, for billions of devices at once.

Function: transfer money to pay for goods. **Capability:** clears a tap in under a second, 24/7, while screening it for fraud against millions of patterns.

The Capability Amplifier

Here's the distinction made physical. The **function stays locked** — this system does exactly one thing: *locate*. The dials below only change its **capability**. Watch the globe respond.

The five-systems tour

VCAA's five example domains, as a rotating hub. Click a pod to focus it — each plays a small animation and reveals its function, its capability, and the **opportunity** a designer could chase.

Help people with disability do tasks they'd otherwise find hard — e.g. a hearing aid amplifies sound, a screen reader voices on-screen text.

Separates speech from noise in real time, auto-tunes per environment, describes images with AI, and streams audio straight to the device.

The five domains, on paper

The same five, in a table you can revise from. Remember: these are **examples** of the function-vs-capability idea, not a memorise-list. The skill is doing this split for whatever system an exam hands you.

Capability is an emergent property

Capabilities don't live in one part — they emerge when the parts work *together*. Switch components off below and watch a navigation system's capability collapse.

How capability grows — and feeds new ideas

Functions are fairly stable (we've "located" and "paid" for a long time). It's the **capability** that keeps climbing — and each climb opens a door for an innovative solution. This is the bridge back to KK01's drivers of innovation.

Cheap sensors and phones pour out data. Traffic systems went from a few loop detectors to whole-city, second-by-second sensing — so signals can now adapt instead of running on a fixed timer.

Always-on networks let parts that were isolated talk live. Your navigation reroutes around a crash *because* thousands of phones report speeds back in real time.

Cheaper compute and machine learning add **autonomy** — fraud detection, speech-from-noise, robots that adapt to what they sense rather than following a fixed script.

What filled a room now fits a wristband. That's exactly why **FloatMate** (KK01) is feasible now — the capability finally fits on the wrist at a price a surf club can afford.

Sort it: function or capability ?

Drag each statement into the right bin — or tap it, then tap a bin. Some are deliberately sneaky (a "benefit" hiding as a capability).

Six named exam traps

These are the recurring ways students bleed marks on this dot point. Name the trap, dodge the trap.

Command-word decoder

The verb in the question sets how much you write. Tap one to see what it's really asking on *this* dot point.

Just name it — no explanation needed.

State the function of a GPS navigation system. → "To locate a device and provide directions."

P·E·L writing trainer

Point → state the function or capability. **Evidence** → back it with a concrete measure or example. **Link** → tie it to the question (often: the opportunity it creates). Type an answer and watch the meters react live.

VCE-style practice questions

Five questions building from 2 to 4 marks. For each: reveal a **weak** answer (annotated with why it loses marks), a **strong** answer, and the **marking guide**. Then mark yourself — your score fills the ring, bottom-right.

"The function is what it does and the capability is also what it does but better."

No real distinction — "what it does but better" is hand-wavy. A distinguish question needs both sides defined against each other, ideally with a contrast word like *whereas*.

"A **function** is the task a digital system is designed to perform — what it is for (e.g. to locate a device), *whereas* a **capability** is the extent of what it is able to do — its scale, speed, accuracy or reach (e.g. locating to within ~3 m anywhere on Earth in real time)."

- 1 Function correctly defined as the task/purpose — what the system does.
- 1 Capability correctly defined as the extent/how well it does it, clearly contrasted with function.

An example for each is not required for 2 marks but strengthens a borderline answer.

"Its function is to take money. It's really fast and it's safe and convenient."

Function is acceptable (1). But "fast", "safe" and "convenient" are vague and "convenient" is a benefit, not a capability — no measures, so the two capabilities don't earn their marks.

Cheat sheet

Function = what it does (one verb, its purpose).

Capability = what it's able to do (a measure: scale/speed/accuracy/reach/autonomy).

- Has a verb only? → function.
- Has a number / "real-time / 24-7 / global"? → capability.
- Describes a user feeling ("handy")? → benefit, not capability.
- Hardware + software + data + network + people.
- Capability is **emergent** — name ≥2 parts working together.

Assistive tech · financial services · GPS · robotics · traffic management. **"For example" = not a memorise-list.** Apply the method to any system.

- More data · more connectivity
- More compute & AI (autonomy)
- Smaller, cheaper hardware

Describe current capability precisely → see the gap between what it can do and what people need → that gap is where the innovative solution lives.

Common traps & misconceptions

Exam trap

TRAP 1 Treating function and capability as the same word ▸ Writing "the function of GPS is that it's very accurate." Accuracy isn't a function — it's a capability. The function is *to locate*. **Fix:** Function = one verb (what it's for). Capability = a measure (how well/much/far). If your sentence has a number or "real-time / city-wide / 24/7" in it, you're describing a capability.

Exam trap

TRAP 2 Listing buttons and features instead of capabilities ▸ "It has a search bar and a settings menu." Those are features, not capabilities. A capability is what the system can *achieve*, stated with scope. **Fix:** Push past the feature to the outcome with a measure — not "it has live traffic" but "it can reroute around congestion in real time using live data from millions of devices."

Exam trap

TRAP 3 Describing one part, not the system ▸ "A robot arm lifts things." That's a component. The *system* is the arm **plus** its sensors, control software, data and the people who program and supervise it. **Fix:** Name at least two cooperating parts (hardware + software/data/network) so it reads as a system, not a gadget.

Exam trap

TRAP 4 Confusing a capability with a benefit ▸ "Its capability is that it's convenient." Convenient is a *benefit to the user*, not something the system does. Benefits are the *result* of a capability. **Fix:** State the capability first ("settles a payment in under a second"), then, if asked, the benefit it gives ("so queues move faster").

Exam trap

TRAP 5 Vague capability with no measure ▸ "It's fast and powerful." Examiners can't award that — it could describe almost anything. **Fix:** Add scale: *how* fast (sub-second), *how* many (millions/sec), *how* accurate (±3 m), *how* far (global, city-wide). The measure is the mark.

Exam trap

TRAP 6 Memorising the five examples as "the answer" ▸ VCAA wrote "for example", so assistive tech / finance / GPS / robotics / traffic are *illustrations*. A question may hand you a totally different system. **Fix:** Practise the *method* — split any system into function + capability — so you can apply it cold to a vending machine, a smart watch or a drone you've never revised.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Distinguish · 2 marks

Between the **function** and a **capability** of a digital system.

✗ A weak answer

— would score 0–1 "The function is what it does and the capability is also what it does but better." No real distinction — "what it does but better" is hand-wavy. A distinguish question needs both sides defined against each other, ideally with a contrast word like *whereas*.

✓ A strong answer

— full marks "A **function** is the task a digital system is designed to perform — what it is for (e.g. to locate a device), *whereas* a **capability** is the extent of what it is able to do — its scale, speed, accuracy or reach (e.g. locating to within ~3 m anywhere on Earth in real time)."

How the marks are awarded

- 1 Function correctly defined as the task/purpose — what the system does.
- 1 Capability correctly defined as the extent/how well it does it, clearly contrasted with function.

Q2. Identify · 3 marks

For a **contactless payment terminal**, identify its main function and describe **two** of its capabilities.

✗ A weak answer

— would score 1 "Its function is to take money. It's really fast and it's safe and convenient." Function is acceptable (1). But "fast", "safe" and "convenient" are vague and "convenient" is a benefit, not a capability — no measures, so the two capabilities don't earn their marks.

✓ A strong answer

— full marks **Function:** to authorise and transfer payment for a purchase between a customer's account and the merchant. **Capability 1:** it can complete and settle a tap in under a second, operating 24/7 without a teller. **Capability 2:** it can screen each transaction against millions of fraud patterns in real time and decline suspicious ones automatically.

How the marks are awarded

- 1 Function: process/authorise a payment between customer and merchant.
- 1 One capability described *with a measure or scope* (e.g. sub-second, 24/7).
- 1 A second, distinct capability, also with scope (e.g. real-time fraud screening).

Q3. Explain · 4 marks

How the capabilities of **GPS-based navigation systems** create opportunities for innovative solutions.

✗ A weak answer

— would score 1–2 "GPS helps you find places. People made Uber with it. So it creates opportunities." States a function ("find places") but never names a *capability* with scope, and the link to innovation is asserted, not explained. "Explain" needs the chain: capability why it enables something the opportunity.

✓ A strong answer

— full marks "GPS navigation can pinpoint a device to within a few metres, anywhere on Earth, in real time and for billions of devices at once. **Because** location is now cheap, live and universal, designers can assume a user's position is always known. **This** makes solutions like real-time ride-share matching, live delivery tracking and geofenced safety alerts feasible — the system supplies precise, continuous location, so the innovation only has to add value on top of it rather than solve locating from scratch."

How the marks are awarded

- 1 Names a genuine capability with scope (precision, real-time, global, scale).
- 1 Explains *why* that capability matters (cheap/live/always-available location).
- 1 Connects it to a concrete opportunity/innovative solution.
- 1 Reasoning is a clear chain (capability enabler opportunity), not just assertion.

Q4. Analyse · 4 marks

Transfer task. A local council wants to cut peak-hour congestion on a busy corridor. **Analyse** how the capabilities of an adaptive traffic-management system could meet this need.

✗ A weak answer

— would score 1–2 "Traffic lights control cars. If they're smart they'll fix the congestion because they're better than normal lights." Names the function (control cars) but the capabilities are missing and the reasoning is circular ("better… so it fixes it"). Analyse wants the capability broken down and weighed against the specific need (peak-hour congestion on one corridor).

✓ A strong answer

— full marks "An adaptive system can **sense traffic volume in real time** across the whole corridor (capability), so instead of fixed timers it can **lengthen green time in the busy direction** during the peak. It can also **give priority to trams and buses**, moving more people per cycle, and **coordinate adjacent intersections** into a 'green wave' to keep flow moving. Matched to the council's need, these directly attack peak-hour congestion on that corridor. A limitation to weigh: the gain depends on accurate sensor data and won't help if the road is simply over capacity — so it manages flow rather than removing the underlying demand."

How the marks are awarded

- 1 Identifies a relevant capability (real-time sensing / adaptive timing).
- 1 A second capability (PT priority or intersection coordination).
- 1 Links capabilities specifically to reducing peak-hour congestion on the corridor.
- 1 Weighs it up — a limitation or condition (data dependence / capacity limit) for true "analyse".